2. Final Report: Medical Aid Cost Prediction Using Linear Regression

2.1 Introduction

The objective of this analysis was to develop a predictive model that estimates medical aid charges based on demographic and lifestyle factors. The model was built using a Multiple Linear Regression approach to support a medical aid scheme in implementing data-driven pricing strategies.

The dataset included variables such as age, sex, BMI, number of children, smoking status, and geographic region, with medical charges as the target variable.

2.2 Data Cleaning and Preparation

A structured data preparation process was conducted to ensure model reliability:

2.2.1 Data Quality Checks

- The dataset was inspected using .info() and .describe()
- No missing values were detected, ensuring completeness
- Data types were validated and confirmed as appropriate

2.2.2 Encoding Categorical Variables

Categorical variables (sex, smoker, region) were converted into numerical format using **one-hot encoding** to ensure compatibility with regression modelling.

2.2.3 Outlier Consideration

- Outliers were observed in the charges variable.
- These values were retained as they represent valid high-cost medical cases (e.g., smokers with chronic conditions).

2.3 Exploratory Data Analysis (EDA)

EDA was conducted to understand data patterns and relationships.

2.3.1 Distribution of Medical Charges

- Charges were found to be **right-skewed**
- Most individuals fall in lower to mid-cost ranges, with a small number of high-cost cases

Insight: This indicates that healthcare costs are not evenly distributed.

2.3.2 Age vs Charges

- A positive relationship was observed.
- Older individuals generally incur higher medical expenses.

Insight: Age is a significant predictor of healthcare cost.

2.3.3 Smoking Status

- Smokers incur significantly higher charges than non-smokers.
    - This was the strongest observable relationship in the dataset

Insight: Smoking is a key risk factor driving insurance costs.

2.3.4 BMI vs Charges

- Moderate positive correlation exists
- Higher BMI is associated with increased medical costs

2.3.5 Correlation Analysis

- Strongest correlation: Smoking status and charges
- Moderate correlation: Age and BMI
- Weak correlation: Region and sex

2.4 Model Training

A Multiple Linear Regression model was used due to its interpretability and suitability for continuous prediction.

2.4.1 Data Preparation for Modelling

- Dataset was split into:
    - 80% training data
    - 20% testing data
- One-hot encoding was applied to categorical variables

2.4.2 Model Implementation

- Algorithm: Multiple Linear Regression
- Library: Scikit-learn
- Hyperparameters: Default settings were used

The model was trained using the training dataset and tested on unseen data to evaluate generalisation performance.

2.5 Model Evaluation

The model was evaluated using multiple statistical metrics:

2.5.1 Performance Metrics

Metric	Purpose
R^2 Score	Measures variance explained by the model
MAE	Average prediction error
MSE	Penalises large errors
RMSE	Provides interpretable error magnitude

2.5.2 Interpretation of Results

- **R^2 Score:** The model explained a substantial proportion of variance in medical charges, indicating good predictive capability.
- **MAE:** Showed that predictions were reasonably close to actual values on average.
- **RMSE:** Indicated some larger errors, likely due to high-cost outliers.

2.5.3 Residual Analysis

- Residual plots showed random dispersion around zero
- This suggests that linear regression assumptions were largely satisfied

2.6 Key Findings

1. Smoking status is the most influential factor in determining medical costs
2. Age and BMI significantly affect insurance charges
3. Geographic region has minimal predictive power
4. Medical costs are highly skewed, with a small group driving high expenditure

2.7 Business Interpretation

The model provides actionable insights for a medical aid scheme:

- Premiums can be adjusted based on **risk profiling**
- Smokers represent a high-cost group requiring pricing adjustments
- Preventative health programmes could reduce long-term costs
- Age and BMI can be used for tiered pricing structures

2.8 Model Limitations

- Dataset is US-based and may not fully represent South African healthcare dynamics
- Linear regression assumes linear relationships, which may oversimplify reality
- External factors such as medical history and income were not included

2.9 Recommendations

- Collect and train on South African-specific data
- Explore advanced models such as Random Forest or Gradient Boosting
- Incorporate behavioural and clinical health indicators
- Deploy model as part of a decision-support system for pricing strategy

2.10 Conclusion

The analysis successfully demonstrated the use of Multiple Linear Regression to predict medical aid charges based on lifestyle and demographic features. The model provides a strong baseline for data-driven pricing strategies and highlights smoking, age, and BMI as key cost drivers.